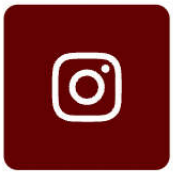
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HY Dermatology

This document is not designed to be a long-winded, 350-page dermatology textbook that caters to superfluous details that will never be tested. The focus here is to be as concise as possible with HY factoids in order to increase your score on the USMLE.

- Student Q showed 10M with scalp lesion similar to below, then the Q asked for the treatment:



  - Answer = oral griseofulvin for patient only (also on FM NBME form); wrong answer = "oral griseofulvin for patient and classmates"; Dx is tinea capitis; note alopecia and circular/scaly appearance of lesion; cause is dermatophytes (i.e., *Microsporum*; *Trichophyton*).
  - Q on different NBME asks how to prevent; answer = "avoidance of sharing of hats"; "use of medicated shampoo" is wrong answer.
- 24M + itchy patches and greasy scales along the hairline; Q asks for the diagnosis:



  - Answer = seborrheic dermatitis (dandruff); treatment = topical selenium or ketoconazole shampoo; does not cause circular area of alopecia as with tinea capitis; more common in adults (tinea capitis more common in children); cause is inflammatory response to over-colonization with *Malassezia* yeast.
  - High prevalence in HIV patients; sudden onset in MSM → answer = do HIV test.

- 46F + has three dogs at home; the following lesion from her forearm is shown; Q asks treatment:



  - Answer = topical miconazole or clotrimazole; diagnosis = tinea corporis (ringworm); Q will often mention dogs or use of yoga mats at the gym.
- 35F + BMI of 55 + type II diabetes + red, moist 8x12-cm ellipse under right breast; Q asks biggest risk factor for her condition? → answer = insulin resistance (increases risk for cutaneous *Candida*); obesity is wrong answer; diagnosis is cutaneous Candida; treat with oral fluconazole.
- 27F + white, cheese-like discharge per vaginum; Q asks what oral treatment she needs; answer = fluconazole; some students say, "Wait, I thought we use topical nystatin" → either oral fluconazole or topical nystatin can be used; there's an NBME Q for Step 1 where they specify "oral" treatment; fluconazole is correct and nystatin isn't listed.
- 32M + fever 101 F + red, itchy, scaly area between his 1st and 2nd toes + the redness/scaling extends up dorsum of foot and onto ankle; Q asks most likely causal organism *for his fever*; answer = *Staph aureus; Trichophyton* is wrong answer; diagnosis is Staph cellulitis superinfection over tinea pedis; Staph can cause the fever; unlikely for tinea pedis in isolation to cause fever.
  - Tx for tinea pedis on USMLE is topical terbinafine or -azole (i.e., clotrimazole/miconazole).
- 40F + diabetic foot ulcer; sterile probe to base of lesion is likely to show what? → correct answer on new NBME exam = "polymicrobial"; wrong answers are *Staph aureus* and *Pseudomonas*. This is an extremely important Q from NBME because people have long debated *Staph* vs *Pseudomonas* for diabetic foot ulcers.
- 6M + puncture wound on foot 3 weeks ago + continues to have warmth, redness, and pain on palpation; Q asks most likely organism (polymicrobial not listed) → answer = *Pseudomonas* on new CMS Peds form; *Staph aureus* is wrong. Apparently implication is osteomyelitis has occurred due to

non-healing / continued pain. Osteomyelitis (bone infection) can occur as a slow/insidious onset and progression. If osteomyelitis occurs from a **plantar** puncture wound, choose *Pseudomonas*.

- 24M + excoriated rash on groin and inner ankle + rash on ankle was successfully treated with topical clotrimazole a few weeks ago, but rash has reappeared + is on groin; what's the mechanism? → answer = "autoinfection" (i.e., he is scratching/re-infecting himself); wrong answers are related to immunodeficiency. Call it stupid, but it's on 2CK NBME form. You need to know "excoriations" mean the patient has been scratching. So if there's excoriations at multiple locations, he's likely scratched / auto-transferred the infection.
- 60M + farmer + thickened yellow nailbed of left big toe; Dx + Tx? → answer = onychomycosis (fungus of nails); Tx = oral terbinafine (if not listed, choose griseofulvin).
- 17F + *Candida* skin infections since childhood + 2-year Hx of type I diabetes mellitus + 1-yr Hx of autoimmune thyroiditis; Q asks mechanism for patient's condition → answer = "deficiency of cell-mediated immunity"; diagnosis is chronic mucocutaneous candidiasis; USMLE wants you to know this is a **T cell problem;** autoimmune conditions go together (i.e., increased risk of one → increased risk of another); this also applies to immunodeficiencies in relation to autoimmunity (e.g., IgA deficiency also associated with atopy and vitiligo); although *Candida* infection risk increases with diabetes, the infections in this patient far precede the diabetes Dx.
- 31M + gardener + has presentation shown below; Q asks the mechanism for this patient's condition:



  - Answer = lymphangitis; diagnosis is lymphocutaneous sporotrichosis (*Sporothrix schenckii*); this image of erythematous lesions streaking toward the axilla is exceedingly HY for lymphangitis; sometimes the image can just be a fine pink streak without open lesions. Wrong answers are phlebitis, arteritis; treatment is oral itraconazole. Students early in their

prep should know that *Sporothrix* is classically papule on the finger caused by rose thorns; this presentation is usually too easy for real USMLE though.

  - Exam can also give *Sporothrix* as guy who goes hiking and scratches his face with a stick → gets papule on the cheek that ruptures into oral cavity + causes draining sinus tract; answer = sporotrichosis; wrong answer = craniofacial *Actinomyces*.

- 19M + plays soccer and goes to beach; has condition in image shown below; what is treatment?



  - Answer = topical selenium; diagnosis is tinea versicolor (*Malassezia furfur*); fungus causes degradation of fatty acids within the skin leading to hypopigmentation; this image is all over the NBMEs for Steps 1 and 2.

- 2-month-old girl + red papules in groin area and intergluteal cleft; family has Hx of asthma; what's the diagnosis? → answer = *Candida* (diaper rash); not atopic dermatitis; the latter can occur in babies but is more often on trunk, dorsa of hands, and face. Treat *Candida* diaper rash with topical -azoles or nystatin.
- 48M + IV drug user + treated for 6 weeks in hospital on broad-spectrum antibiotics; intertriginous red rash is seen; organisms are cultured as purple-budding organisms; diagnosis? → answer = *Candida*; broad-spectrum antibiotics → increased risk of *Candida* infections.
- 42M + fever 100.8 F + diffuse, pink lesion shown on leg below; Q asks most appropriate treatment:



- Answer = oral dicloxacillin, cephalexin, or Augmentin (amoxicillin + clavulanate); diagnosis is cellulitis (infection of the dermis and hypodermis [subcutaneous fat]); *Staph aureus* exceeds *Strep pyogenes* (Group A Strep) as causal organism; must give beta-lactamase-resistant beta-lactam in the methicillin class (i.e., dicloxacillin, flucloxacillin) or first-generation cephalosporin (i.e., cephalexin, cephazolin), or Augmentin (amoxicillin-**clavulanate**); amoxicillin and penicillin alone are wrong answers; 90% of community *Staph* (i.e., MSSA) produces beta-lactamase, so amoxicillin and penicillin alone will not work if *Staph* is the cause.

- 35M + fever 100.5 F + leg has lesion shown + Q asks most likely causal organism:



  - Answer on NBME = *Strep pyogenes* (Group A Strep); diagnosis is erysipelas (infection of upper dermis and superficial lymphatics); Group A Strep eclipses *Staph aureus* for erysipelas; looks worse than cellulitis but is more superficial / "not as bad"; has characteristic "fiery red" appearance and may appear well-demarcated with raised edges. Although Group A Strep > *Staph* for erysipelas, Tx is same as cellulitis (oral dicloxacillin, cephalexin, or Augmentin) because Staph can still cause it. Penicillin alone can be used for *Strep* pharyngitis.

- 7M + presentation shown; Q asks for the treatment:



  - Answer = topical mupirocin; diagnosis is impetigo (school sores); *Staph aureus* exceeds Group A Strep for both bullous *and* non-bullous types (bullous generally implies *Staph*); if orals given, use dicloxacillin or cephalexin, but USMLE loves topical mupirocin for impetigo.

- 16M + cellulitis + BP of 80/40; Q asks which immunologic receptor(s) is/are bound in this patient's condition → answer = MHC-II and T-cell receptor; diagnosis is toxic shock-like syndrome caused by exotoxin A (erythrogenic toxin) of *Strep pyogenes* (Group A Strep); mechanism is similar to *Staph aureus* superantigen, TSST, which bridges MHC-II on macrophages and TCR, causing cytokine release from macrophages.
  - Toxic shock syndrome Qs will mention low BP in someone with cotton nasal packing or tampons → answer is "MHC-II and TCR" bound by TSST of *Staph aureus*.
  - In contrast, cellulitis causing shock, the answer will be *Strep pyogenes* (Staph will not be listed) due to its exotoxin A.
- Neonate + diffuse pink body rash + desquamation of palms and soles; Q asks for molecular target of the toxin in this condition; answer = desmosomes (hold adjacent keratinocytes together); diagnosis is *Staphylococcal* scalded skin syndrome; epidermolytic exotoxin cleaves desmoglien-1 in desmosomes; (+) Nikolsky sign (sloughing of the skin with friction).
- 12F + fever + sore throat + red tongue + pink maculopapular body rash; Dx + Tx? → scarlet fever caused by *Strep pyogenes*; presents with "strawberry tongue" and salmon-pink body rash; Tx with penicillin to prevent rheumatic heart disease.
- 14M + fever + rapid, irregular, jerking movements of limbs + following rash on legs as shown; Dx + Tx?



    - Answer = erythema marginatum as seen in rheumatic fever; treatment is penicillin; erythema marginatum is annular (ring-like) and serpiginous (serpent-/snake-like); movements are Sydenham chorea (autoimmune attack against basal ganglia).

- 10M + "yellow crusties" on his forearm for the past week + red urine; what's the diagnosis? → answer = post-streptococcal glomerulonephritis (PSGN) secondary to *Strep* impetigo. USMLE can sometimes write answer as "proliferative glomerulonephritis" or "acute glomerulonephritis." This is a type-III hypersensitivity with immune complex deposition in the kidney. USMLE loves PSGN due to cutaneous *Strep* (i.e., PSGN need not be caused by *Strep* pharyngitis).
    - Proliferative glomerulonephritis = PSGN. Do not confuse this with diffuse proliferative glomerulonephritis (DPGN, which is red urine in SLE), or membranoproliferative glomerulonephritis (MPGN, which is red urine in HepC or malignancy).
- 19M + burned leg playing with firecrackers + while in hospital develops infection of burn site that has a yellow color; Dx? → answer = *Staph aureus* (golden staph); wrong answer is *Pseudomonas*. Although this is HY organism in burns patients, it produces a blue-green color due to pyocyanin, not a yellow color, which is instead golden *Staph*. So USMLE likes to be tricky that way.
- 25F + breastfeeding + red, cracked, fissured nipple; Dx + Tx? → answer = mastitis; usually caused by *Staph aureus*; Tx is oral dicloxacillin + continue breastfeeding through the affected breast.
- 25F + not breastfeeding + upper, outer quadrant non-fluctuant, warm, tender, red mass; Dx? → answer = mastitis; wrong answer is abscess; mastitis = non-fluctuant; abscess = fluctuant; this is on 2CK obgyn CMS form; mastitis need not affect the nipple in breastfeeding woman.
- 25F + recently stopped breastfeeding + tender, fluctuant mass lateral to the nipple; patient is afebrile; mass is not warm or red; Dx? → answer = galactocele (milk retention cyst); can sometimes be confused with abscess since it is fluctuant and can be tender; if abscess, however, they will say red/warm +/- fever.
- 42F + inverted nipple + patient is worried because family Hx of breast cancer; Dx? → answer = ductal ectasia; benign condition; as name implies, simply dilation/widening of lactiferous duct.
- 65F + red, eczematoid-appearing nipple + mass palpable beneath nipple; Dx? → answer = Paget disease of breast; often associated with underlying ductal carcinoma *in situ*.

- 65F + peau d'orange of left breast + erythematous; Dx? → answer = inflammatory breast cancer; peau d'orange appearance due to Coopers ligaments of breast (on NBME).
- 17M + presents as per image shown; what's the diagnosis? (answers are either *Propionibacterium acnes* or tinea faciei):



  - Answer = acne; *Propionibacterium acnes*; not difficult, but I've seen enough students select tinea faciei.
  - First-line Tx for acne on USMLE is topical retinoids (i.e., topical tretinoin; **not** oral isotretinoin; latter is only for severe acne). Topical retinoids (vitamin A) inhibit sebum production; they cause photosensitivity and desquamation (peeling).
  - Topical benzoyl peroxide is second-line for acne (although often co-administered with topical tretinoin). It clears pores and kills bacteria.
  - Topical clindamycin can be used if topical retinoids and benzoyl peroxide are insufficient; if topical antibiotic is insufficient, oral tetracycline is used; the latter causes blistering photosensitivity.
  - Last resort is oral isotretinoin; must do beta-hCG (pregnancy test) before commencement due to teratogenicity; oral isotretinoin does **not** cause problems with sperm in men; topical retinoids in both men and women do not cause teratogenicity.
- 20F + being treated for acne with both topical + oral medications; her forehead is shown below; what is the most appropriate recommendation for this patient?



- Answer = “avoidance of sun exposure”; above rash is classic photosensitivity caused by topical retinoids; tetracycline photosensitivity tends to be blistering; do not choose answers such as “avoidance of spicy/sweet foods” for acne questions.

- 16M + face shown in following image; Dx + Tx?



- Answer = Dx is tinea faciei; Tx = topical -azoles (clotrimazole, miconazole).

- 3M + fever of 103 F + stiff neck + low BP; following image of patient’s leg is shown. What is the most likely causal organism?



- Answer = gram-negative diplococci; Dx is meningitis + characteristic non-blanching rash caused by *Neisseria meningitidis.* Low BP can be endotoxic shock, but student should bear in mind Waterhouse-Friderichsen syndrome is often asked; give hydrocortisone to increase BP

after normal saline is administered. WHF syndrome = bilateral hemorrhagic necrosis of the adrenal cortices caused by meningococcal septicemia → low aldosterone + low cortisol.

- 23M + 2-day Hx of soreness of left knee and right elbow + positive Finkelstein test on right hand + cutaneous papules visualized on right wrist; Dx? → answer = gonococcal arthritis; will present one of two ways on USMLE; 1) monoarthritis of large joint, such as the knee; 2) polyarthritis + tenosynovitis (e.g., deQuervain) + cutaneous papules/vesicopustules. The USMLE will sometimes just have "gram-negative diplococcus" as the answer. Finkelstein test is an MSK exam maneuver to diagnose deQuervain tenosynovitis.
- 28F + recently immigrated to US from India + rose spots on abdomen + severe constipation + fever 104 F + question asks how this condition is acquired; answer = "ingestion of fecal-contaminated food"; diagnosis is typhoid (*Salmonella typhi*); humans are the reservoir; classically causes rose spots on the abdomen + prostration (patient is lying supine + in pain) + either constipation or diarrhea; do not confuse with the food poisoning Salmonella species (*typhimurium* and *enteritidis*), which classically are acquired from eggs/poultry, or reptiles (USMLE likes turtles and lizards).
- 40F + penetrating trauma to thigh one week ago + skin has black appearance and crepitus on exam; the most likely causal organism can also cause what? → answer = watery diarrhea; Dx is gas gangrene due to *Clostridium perfringens,* resulting in subcutaneous emphysema (crunching of skin due to underlying CO_2 gas); this is due to production of lecithinase (phospholipase); *C. perfringens* also causes watery diarrhea.
- 56F + poorly controlled diabetes + *Pseudomonal* sepsis + following image is shown; Dx?



  - Answer = ecthyma gangrenosum; uncommon cutaneous infection seen in patients with severe *Pseudomonal* infections.

- 23F + works as postal worker + horticulturist + has many pet birds; following image is shown + Q describes lesion as an eschar with surrounding edema; Dx?



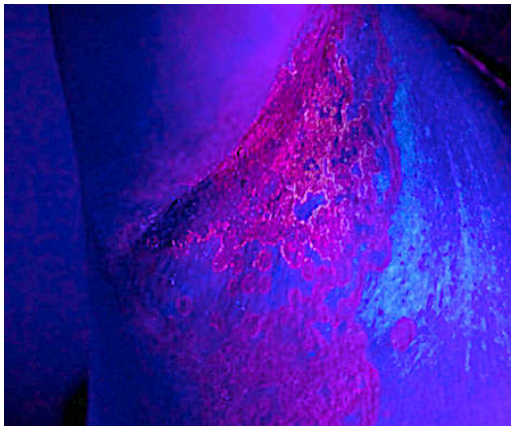
  - Answer = *Bacillus anthracis* (anthrax); cutaneous anthrax can present classically with a black, eschar lesion; the gram (+) rod increases cAMP and produces edema factor.

- 28F gives birth to stillborn neonate + she ate soft cheeses and deli meat while pregnant + stillborn has diffuse granulomas on body; which of the following best describes the most likely causal organism? → answer = gram (+) rods; *Listeria* can cause granulomatosis infantiseptica (severe intrauterine infection often resulting in fetal demise + diffuse cutaneous granulomas).
- 39M + bilateral pneumonia + skin ulcer on back of hand (image shown) + low-grade fever + has many pet rabbits; what's the diagnosis?



  - Answer = *Francisella tularensis;* cutaneous tularemia can present as ulcerative lesions; can also cause bilateral atypical pneumonia; rabbits are classic source.

- 54M + type II diabetic + red rash under right axilla + rash appears bright coral red under Woods lamp (image shown); Dx?



- Answer = erythrasma, caused by *Corynebacterium minutissimum*; skin infection of intertriginous areas that shines/glows coral red (holy shit) under Woods lamp.

- 40M + painful erythematous lesions in her axillae; an image is shown below; what's the best treatment for this patient?



- NBME answer = "surgical excision of lesions"; diagnosis is hidradenitis suppurativa, a skin condition characterized by painful abscesses and sinus tracts, most commonly in the groin, axilla, and under the breasts; frequently affects apocrine glands; cause is a combination of genetic and environmental factors (idiopathic).

- 17M + athlete + strong body odor when finished with sports; Q asks what type of gland is responsible → answer = apocrine; Types of glands:
  - USMLE wants you to know apocrine glands are found in the groin and axilla and are responsible for body odor. Part of the plasma membrane buds off with the substance secreted by the cell. "Apocrine metaplasia," is a term sometimes applied to fibrocystic change of the breast (as well as "blue dome cysts" and "sclerosing adenosis").

  - Eccrine (merocrine) glands are simple sweat glands found all over the body. The secreted substance is merely exocytosed from the cell.
  - Sebaceous glands (holocrine) are those attached to hair follicles (not found on palms/soles). The cell disintegrates when it secretes the substance.

- 34F + diabetic + frequently uses hot tub + image shown below; Dx?



  - Answer = hot tub folliculitis; most frequently caused by *Pseudomonas*; increased risk of *Pseudomonal* infections in diabetics (e.g., otitis externa).
    - Folliculitis = inflammation around hair follicle.
    - Furuncle (boil) = abscess involving single hair follicle (usually *Staph aureus*).
    - Carbuncle = cluster of boils (involving 2+ hair follicles) coalescing into one cutaneous cavity of pus.
    - Abscess = collection of pus not specifically involving hair follicle.
    - For 2CK surg Qs, drain cutaneous pus collections *then leave open to the air* by stuffing with sterile gauze + pulling out slowly daily.

- 62M + poorly controlled diabetes + black skin on perineum + patient is hemodynamically stable and ABCs addressed; what's the next best step in management? → answer = debridement of necrotic tissue; diagnosis is Fournier gangrene (rare perineal/scrotal gangrene seen in older male diabetics); *Clostridium perfringens* is most common culprit.
- 43F + recent penetrating trauma to leg + necrotic tissue that on examination appears to spread along fascial planes; what's the most likely diagnosis? → answer = necrotizing fasciitis; Tx = IV antibiotics + debridement of necrotic tissue.

- 27F + painful 1-cm lump in labia majora; what's the most likely causal organism? → answer = polymicrobial; diagnosis is Bartholin gland cyst/abscess; uncomplicated cysts → treatment = warm compresses or sitz bath; overt abscess → Tx answer = drainage.
- 8M + fever for 5 days + palms and soles desquamation + edema of dorsa of hands + cervical lymphadenopathy + injection of conjunctiva and lips; Dx + Tx? → answer = Kawasaki disease; **5+ days** of fever is HY; Tx = aspirin + IVIG (never give aspirin to kids for other purposes because of Reye syndrome).
- 24F + lives in Connecticut + went hiking five days ago + rash of wrists and ankles + rash migrates in toward chest; Dx + Tx? → answer = Rocky Mountain spotted fever (*Rickettsia rickettsii*); classically palms and soles rash, but USMLE also sometimes says wrists + ankles; student should be aware rash is centripetal (i.e., starts at palms/soles, or wrists/ankles, and moves inward to trunk); treatment is doxycycline for anyone age 9 and older; pregnant women and children 8 and younger receive other agents, such as amoxicillin or ceftriaxone.
- 6M + vesicular eruption on bottoms of feet, palms, and periorally; most likely causal organism? → answer = "RNA virus; non-enveloped; non-segmented"; diagnosis is hand-foot-mouth disease caused by Coxsackie A virus; this virus can also cause herpangina (posterior oropharyngeal vesicles).
- 24M + recently returned home from military service + fever of 101 F + cervical lymphadenopathy + posterior oropharyngeal vesicles + painful vesicles inside the lip (nothing on outside of mouth/lips); Q asks next best step in diagnosis? → answer = "PCR testing of vesicles"; viral culture is wrong answer; diagnosis is HSV1/2; do not confuse with herpangina; HSV primary infection will classically present with fever and lymphadenopathy.
- 54F + image shown below; Q asks most likely organism in terms of viral structure (i.e., DNA vs RNA; enveloped vs non-enveloped; circular vs linear):



- Answer = DNA, enveloped, linear; Dx is HSV1/2; herpes labialis.

- 40F + works as dentist + image as shown below; what's the MOA of the treatment?



- Treatment = DNA polymerase inhibitor (causes "chain termination"); acyclovir (or valacyclovir); Dx is herpetic whitlow, which is an HSV1/2 infection of the finger, often caused by inoculation from a cold sore; increased prevalence in dentists/hygienists.

- 40M + gardener + photo is shown; this is due to what type of toxicity?



- NBME answer = arsenic; Mees lines are white lines seen on fingernails in arsenic toxicity; arsenic is present in small amounts in fertilizers, which causes plants to flourish; gardeners + landscapers at increased risk; arsenic can also cause palms/soles rash (arsenical keratosis).

- 8M + recent paranasal sinus infection + now presents with fever 103 F + painful, swollen eye as shown below; Dx + Tx?



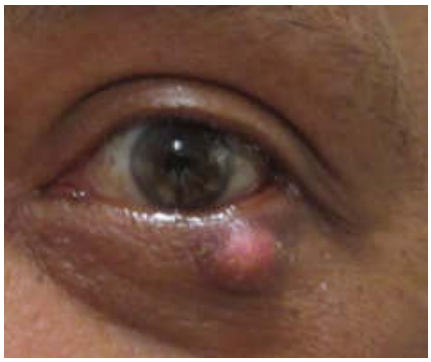
  - Dx is orbital cellulitis (infection involving tissues posterior to orbital septum); *Staph aureus* most common cause; rare sequela of adjacent infection, e.g., from the paranasal sinus; medical emergency; requires IV antibiotics; preseptal cellulitis is a less severe version of orbital cellulitis.

- 30F + painful, red bump on upper eyelid; image shown; Dx + Tx?



  - Dx is hordeolum (stye); *Staph aureus* infection of sweat gland or oil duct; treat with warm compresses.

- 44M + painless bump on eyelid; image shown; Dx + Tx?



  - Dx is chalazion; blocked oil duct; not an infection; treat with warm compresses. 2CK in particular likes hordeolum vs chalazion.

- 24M + painless ulcer on his penile shaft; Dx + Tx? → answer = primary syphilis (chancre sore described); organism is *Treponema pallidum*; diagnose with darkfield microscopy to visualize spirochetes; Tx with penicillin.
  - Do not confuse with chancroid, which is a painful ulcer caused by *Haemophilus ducreyi*.

- 24M + Hx of unprotected intercourse + rash as shown in image below; KOH prep is negative; Q asks what is most likely to confirm diagnosis:



- Answer on USMLE = fluorescent treponemal antibody(FTA); Dx is secondary syphilis showing characteristic maculopapular/nodular body rash; negative KOH means it's not tinea versicolor; palms and soles are classically affected in secondary syphilis but Q need not mention it as per above. It's the negative KOH where you say, "Well if it's not tinea versicolor, what other DDx could it be? "Secondary syphilis is diagnosed with serology; Q need not mention Hx of chancre from primary syphilis.
- VDRL/RPR ordered before FTA, but latter more specific. Former can be falsely positive in patients with antiphospholipid syndrome (classically SLE patients with lupus anticoagulant).
- USMLE Q can also show you picture of wart-like lesions on the genitals (condylomata lata) + tell you there's palms/soles rash, then answer = *Treponema pallidum;* this is also secondary syphilis presentation.

- 24M + Hx of unprotected intercourse + recently immigrated to US from Egypt + presents with below image of lesion on forehead + 2/6 decrescendo holo-diastolic murmur auscultated on exam; the most likely causal organism bears taxonomy most similar to which of the following:



- Answer = *Leptospira interrogans*; diagnosis in image above is gumma of tertiary syphilis; syphilis is a spirochete, as are *Leptospira, Borellia burgdorferi* (Lyme disease), and *Borrelia recurrentis* (relapsing fever). Tertiary syphilis can cause ascending aortitis (murmur is AR).

- 7M + from Massachusetts + family has pet dog + has rash as shown below; Dx + Tx?



- Answer = Lyme disease; rash is erythema chronicum migrans (classic target rash); **the rash need not be a target on USMLE; it can merely be circular with no clearing;** but the target is classic; USMLE can give two side by side images: 1) circular rash on limb that is not a target; 2) Bells palsy → student needs to infer this is Lyme disease even though rash isn't a target. Treatment in this kid is amoxicillin, not doxycycline.
- Treatment is doxycycline for most cases of Lyme; doxycycline is not given to children age 8 and younger or to pregnant women (causes teeth discoloration); if pregnant or age 8 and younger, give amoxicillin.
- Be aware ceftriaxone can be given for severe Lyme that is disseminated, causing cardiac or cognitive dysfunction.

- 24F + pregnant + Bells palsy + target rash; Q asks for treatment (answers are steroids, doxycycline, ceftriaxone) → answer = ceftriaxone; doxycycline not given to pregnant women or children under age 8; student says, "Wait, but I thought you just said ceftriaxone is for severe Lyme; her presentation is simple" → Yes, I agree, ceftriaxone is classically for severe Lyme, but in this Q, they force you to choose it because we can't give doxy; you need to be flexible in some cases (example on NBME).
- 25M + sleeps with pet dog + tick found on the dog + circular rash on arm + blood smear shows phagocytes with intracellular berry cluster organisms; Dx + Tx? → answer = Ehrlichiosis (*Ehrlichia chaffeensis*); bacterium spread by Ixodes tick (same as Lyme disease, Babesia, and Anaplasma); will not cause target rash; "berry cluster organisms" or intracellular "morulae" may be seen.
- 40M + living in homeless shelter for past 4 months + itchy hands + image shown below + topical antifungals were attempted but not effective; Q asks for the treatment:



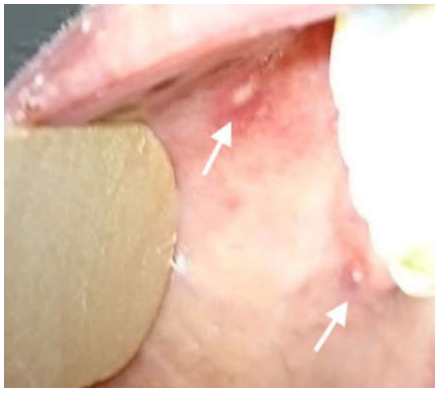
- Answer = topical permethrin; Dx is scabies (*Sarcoptes scabiei*); lesions classically described as "linear burrows"; can become superinfected with *Staph aureus.*
- Disseminated scabies can occur in HIV patients; Tx is oral ivermectin.
- Topical permethrin is also treatment for pediculosis (lice).

- 19F + recent travel through southern US + stayed at dodgy AF motel on the side of the highway next to a Denny's and a Sonic + itchy lesions on arm as shown in image below; Q asks for organism:



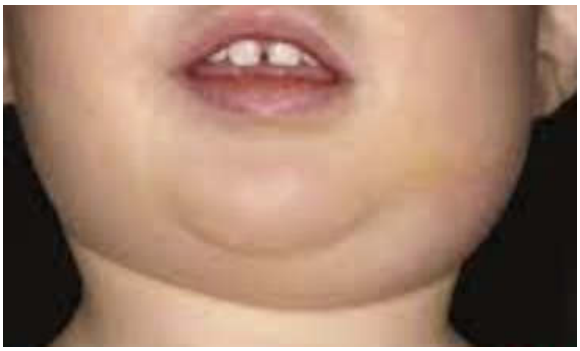
- Answer on NBME = bed bugs; caused by an insect called *Cimex*.

- 6M + family recently immigrated to US from Albania + diffuse maculopapular rash + photograph of buccal mucosa is shown below; Dx + Tx?



- Dx = measles (rubeola); Tx is supportive; image shows Koplik spots (pathognomonic whiteish lesions on buccal mucosa); immigrant Hx on USMLE sometimes implies unvaccinated status.

- 6M + diffuse maculopapular rash + tenderness at base of occiput and behind the ears; Dx + Tx? → Dx is rubella (German measles); Tx is supportive; suboccipital and post-auricular lymphadenopathy are characteristic.
- 4M + fever + image shown below; what is the most likely causal organism (DNA vs RNA; enveloped vs non-enveloped; segmented vs non-segmented)?



  - Dx is mumps → causes POM → Parotitis, Orchitis, Meningitis; virus is RNA, enveloped, non-segmented.
- Neonate + intracranial calcifications + hepatomegaly + rash as shown in image below; no murmurs; Q asks most likely causal organism:



  - Answer = congenital cytomegalovirus (CMV); image shows blueberry muffin rash; no murmurs implies not rubella (causes patent ductus arteriosus).
- 5M + fever 2-3 days ago + brought in by mother to emergency with appearance as shown below; the most likely causal organism can also cause what?



- Answer = aplastic anemia; Dx = Parvovirus B19 (Fifth disease); image shows classic “slapped cheek” appearance; next best step in Dx = Parvovirus B19 IgM titers. If IgM titers are not listed and the child has aplastic anemia, choose bone marrow biopsy. If the patient doesn't have aplastic anemia, you don't need to do the biopsy. There is increased risk of aplastic anemia in sickle cell, but not mandatory the child has that. Once child has developed the red cheeks, he/she has immunologically cleared the illness (i.e., if they turn it into a communications style Q, tell parents to relax because the child has cleared the virus).

- 26F + works at daycare center + fever + rash as shown in image below; next best step in diagnosis?



- Answer = Parvovirus IgM titers. Viral infections in general can present with exanthem (rash). If they say "daycare center," this is hyper buzzy for Parvo on USMLE.

- 17M + high school wrestler + Hx of atopy + has burning, painful rash on trunk + lymphadenopathy + fever; image shown below; Q asks what's the treatment?



  - Answer = oral acyclovir (or valacyclovir); diagnosis is eczema herpeticum → HSV1/2 infection superimposed on eczema; often self-inoculated from touching cold sore then cracked skin of eczema; herpes infection can present with fever, lymphadenopathy, and tingling, burning, and painful rash (herpetic neuralgia).

- 3M + undergoing chemotherapy for ALL + has rash as shown below; what's the diagnosis?



  - Answer = shingles; yes, pediatric shingles "is a thing"; patient is immunocompromised due to chemotherapy.
  - Shingles is aka "herpes zoster" and is caused by varicella zoster virus (VZV); herpes zoster is **not** a virus name; herpes zoster literally is just another name for shingles, which is caused by VZV (human herpes virus 3; HHV3), not HSV1/2 (HHV1/2).

- 84M + sudden-onset left-sided Bells palsy + lesions around the ear as shown below; what's the diagnosis?



  - Answer = herpes zoster oticus (VZV); shingles of facial nerve; can be associated with Bells palsy.
  - Be aware shingles lesions can appear **non-vesicular and black** on USMLE.
  - New FM form for 2CK wants you to know shingles vaccine indicated at age 50.
- 18F + immigrated to US from Mexico in elementary school + presents with crops of pruritic vesicles on the trunk at different stages of healing; Dx? → answer = varicella (chickenpox); immigrant status implies unvaccinated status; "crops of vesicles at different stages of healing" = buzzy for VZV.
- 2M + fever of 103-4 F for 3 days; fever abruptly subsides, followed by rash as shown below; patient's vaccination Hx is up to date; what's the diagnosis?



  - Answer = roseola (HHV6); described as "spiking fever followed by a rash"; child will have high fever for 2-3 days, followed by a rash.
- 22F + no past medical history + afebrile + itchy rash on back for past week as shown below; diagnosis?



- Dx = pityriasis rosea; caused by HHV6 or 7; self-limiting; starts as Herald patch (larger pink ellipse above), usually on the back or trunk, then spreads upward onto the shoulder blades ("Christmas tree distribution"); USMLE will show you image and expect you can make spot-diagnosis.

- 50M + Hx of IV drug use + violaceous skin lesions as shown below; what's the diagnosis?



- Answer = Kaposi sarcoma; usually caused by HHV8 (Kaposi sarcoma-associated herpes virus) in immunocompromised patients (i.e., AIDS, chemotherapy); Kaposi sarcoma presents as tumorous lesions of vascular-lymphatic origin.

- 12M + undergoing chemotherapy for ALL + skin lesions as shown in image below; Q asks you to pick the organism that causes it, but you see answers are all bacteria (i.e., HHV8 not listed):



  - Answer = *Bartonella henselae*; Dx is bacillary angiomatosis; bacterium causes cat scratch disease but is also known to cause Kaposi-sarcoma like lesions in immunocompromised patients.

- 7F + has pet cat + papules on hand + lymph node biopsy shows granulomatous inflammation + organism can be visualized using silver stain; Dx? → answer = cat scratch disease.
- 6M + went to pool party a week ago + has lesions on trunk as shown in image below; what's the most likely causal organism?"



  - Answer = poxvirus (molloscum contagiosum); largest DNA virus; classically skin-colored or reddish papules with central umbilication. Classically acquired from fomites (inanimate objects like metal railings of a pool).

- 9M + comes to your clinic in Brazil + black ulcer on arm + vision loss + patient is successfully treated with ivermectin; what's the most likely diagnosis? → answer = Onchocerciasis (*Onchocerca volvulus*); nematode (roundworm helminth) transmitted by black fly; second most common cause of blindness worldwide after trachoma (Chlamydia A-C).
- 9M + lives in South America + swollen eyebrow + shortness of breath on exertion + S3 heart sound on examination; what's the most likely organism? → answer = *Trypanasoma cruzi* (Chagas disease); protozoan; unicellular eukaryote; spread by Reduviid bug ("kissing bug" because bite is painless); can cause dilated cardiomyopathy and achalasia; swollen eyebrow is referred to as Romaña sign.
- 40F + traveled to Middle East for one month + has lesion on hand as shown below + elevated liver enzymes + pancytopenia; the organism is transmitted by sandfly; what's the most likely diagnosis?



- Answer = leishmaniasis; *Leishmania donovani*; protozoan; unicellular eukaryote; can cause visceral disease known as kala azar, which is associated with ↑ LFTs and pancytopenia.

- 39M + works at aquarium + presents with lesions as shown below; what's the diagnosis?



- Answer = *Mycobacterium marinum*; causes red ulcers/blisters on hand/arm of those who are exposed to aquatics. Do not choose *Staph aureus* if they specifically say aquarium / waterpark in the question.

- 7M + received renal transplant last year + has preauricular reddish lesion shown below; Dx?



- Answer on USMLE = *Mycobacterium avium intracellulare* (MAI); can cause lymphadenitis with classic reddish/violaceous lesion on the neck or preauricularly; MAI classically causes lung disease in older women (Lady Windermere syndrome) and AIDS patients.

- 45M + recently immigrated to US from Libya with adult children + hypoesthesia of hands to pain and temperature + nodularity of fingers and nose; the most likely causal organism most likely bears what characteristic? → answer = "temperature sensitivity"; Dx = leprosy (*Mycobacterium leprae*); grows at cooler temperatures, so can affect the peripheral nerves and nose → can cause neuropathy and disfiguration of face (leonine facies) and limbs.
- 17M + Hx of Celiac disease + lesions on elbows as shown in below photo; what is most likely to be seen on skin biopsy?



  - Answer = IgA deposition at dermal papillae; diagnosis is dermatitis herpetiformis (not actual herpes, in contrast to eczema herpeticum; don't confuse these conditions); this is a cutaneous eruption that can occur in patients with Celiac.

- 35M + Hx of ulcerative colitis + lesion on forearm as shown in below photo; what's the diagnosis?



  - Answer = pyoderma gangrenosum; described as ulcer with necrotic debris; rare cutaneous manifestation in IBD (usually ulcerative colitis).

- 32F + African-American + high serum calcium + image shown + biopsy of lung tissue shows non-caseating granulomas; diagnosis?



  - Image is erythema nodosum. Patient has sarcoidosis. Erythema nodosum is a panniculitis (inflammation of subcutaneous fat); can be caused by autoimmune diseases like sarcoidosis and Crohn, as well as part of serum sickness due to medications (e.g., sulfa).

- 16M + episodes of bloody stool over past two years + perianal abscesses/fistulae seen on physical examination; Dx? → Crohn disease; increased risk since transmural inflammation in Crohn.
- 46M + pain during defecation + perianal skin tag visualized on physical exam + patient won't allow rectal exam due to exquisite pain; Dx + Tx? → anal fissure; usually posterior in the midline; NBME answer is Sitz bath for treatment.
- 16M + painful 2-cm mass located at superior aspect of gluteal cleft; Dx + Tx? → answer = pilonidal cyst/abscess; Tx = surgical closure.
- 20F + receives IM ceftriaxone + oral azithromycin for pelvic inflammatory disease; 3-5 days later she has red rash in area of injection; diagnosis? → answer = Arthus reaction; type III hypersensitivity; can be distinguished from type I (immediate) because type III takes a few days to appear.
  - 20F + receives IM ceftriaxone + gets polyarthritis a few days later; Dx? → answer = serum sickness; type III hypersensitivity.
  - Arthus reaction = redness at injection site 3-5 days later. Serum sickness = polyarthritis 3-5 days after injection, drug, or acute infection. Both are type III hypersensitivites.
- 34F with image shown below; Q asks which other condition is most immunologically similar:



- Image shows malar rash of SLE; answer = serum sickness, Arthus reaction, or PSGN; all four presentations are type III hypersensitivities (the correct answer is any type III hypersensitivity listed in Q answer choices; the point is to know malar rash is due to immune complexes, as are serum sickness, Arthus reaction, and PSGN).

- 52F + increased serum creatine kinase + 3/5 strength of hips on physical exam; images as shown below; what type of antibodies are classically seen in this condition?





- Answer = Anti-Jo1 antibodies; diagnosis is dermatomyositis; left image shows heliotrope rash; often described as violaceous eyelids; **do not confuse with malar rash of SLE;** right image shows shawl sign/rash; both dermatomyositis and polymyositis can have increased serum CK and/or proximal muscle weakness on physical examination (in contrast, polymyalgia rheumatica will have neither, and most often is just pain + stiffness). Muscle biopsy in polymyositis/dermatomyositis shows T cell infiltrate.

- 50F + proximal muscle weakness + image shown below; Q asks for diagnosis:



  - Dx = dermatomyositis; image shows Gottron papules, which is a HY cutaneous finding; patients can also have "mechanics hands," which are rough-surfaced / scaly-appearing hands; do not go chasing fungal infections here.

- 6F + fever 102 F + pain, erythema, and warmth of left knee + Hx of several episodes of joint pains past few years + Hb of 10.1 g/dL + MCV 72; diagnosis? → answer = juvenile idiopathic arthritis (JIA), aka juvenile rheumatoid arthritis (JRA); HY for Peds; patients are susceptible to septic arthritis (as with this patient), but will often have Hx of several episodes of non-septic joint pain; anemia of chronic disease common (MCV can be *low;* I've seen this on multiple 2CK NBME/CMS Qs, where MCV need not be normal); salmon pink body rash classic, but only seen in maybe one-third of NBME/CMS Qs, as per my observation.
- 25F + hands shown in image below; Q asks melanocyte # and melanin production in following condition (i.e., ↑, ↓, or no change):



  - Answer = ↓ melanocyte #; ↓ melanin production; diagnosis is vitiligo; T cell-mediated destruction of melanocytes; can be associated with other autoimmune polyglandular syndromes, as well as IgA deficiency.

- Neonate + milky white skin + blonde hair + pale blue eyes; siblings and parents have darker complexion; Q asks for melanocyte # and melanin production; answer = **normal** melanocyte #; ↓ melanin production; diagnosis is albinism; can be associated with many conditions, including PKU, Chediak-Higashi, and tyrosinase deficiency.
- 18F + pale complexion + freckles on face; Q asks for melanocyte # and melanin production of the freckles; answer = normal melanocyte #, ↑ melanin production; medical term for freckle is ephelis (plural = ephelides).
- 69F + lesion on zygoma as shown below; Q asks for for melanocyte # and melanin production:



- Answer = ↑ melanocyte #; no change melanin production; diagnosis is lentigo (age spot; plural = lentigines).

- 34M + dark complexion + skin sample mixed up in lab with fair-skinned individual; Q asks melanocyte # and melanosome # in dark-skinned individual → answer = no change melanocyte #; ↑ melanosome #; melanosomes are organelles within melanocytes that produce melanin. Differences in skin pigmentation between individuals is due to more or less melanin, while melanocyte number is the same for all people.
- 25F + Hx of gastroesophageal reflux + hands shown in image below; Q asks for what condition this patient is most likely to develop:



- Answer = pulmonary hypertension (secondary to pulmonary fibrosis); diagnosis is limited scleroderma (CREST syndrome); left image shows Raynaud phenomenon, which is color change due to vascular spasm and reactive hyperemia; the right image shows sclerodactyly, which is tightening of skin of the digits.
- Raynaud phenomenon is not limited to scleroderma and can be seen in conditions such as SLE, hyperviscosity syndrome, or can be familial (erythromelalgia).

- 42F + dysphagia + cracked corners of the mouth; image of nail shown below; Q asks for what is most likely to be seen in this patient:



- Answer = low MCV; Dx is Plummer-Vinson syndrome, which is the triad of 1) esophageal webs (dysphagia), 2) angular cheilosis (cracked corners of mouth), and 3) iron deficiency anemia (severe cases can present with koilonychia [spoon-shaped nails] or pica [eating ice, clay, starch]).

- 16F + blood in stool + arthritis + erythematous/silvery scaling lesions on forehead, above upper lip, and on elbows; attempting to remove one of the skin lesions causes bleeding; diagnosis?



- Answer = psoriasis; patient here also has IBD; HLA-B27 sometimes associated (PAIR → Psoriasis, Ankylosing spondylitis, IBD, Reactive arthritis); Auspitz sign is bleeding of psoriatic scales with attempted removal; psoriatic lesions classically on extensor areas but can be on face/forehead; Munro microabscesses are collections of neutrophils in the skin in psoriasis; psoriatic arthritis in some patients shows “pencil-in-cup” deformity on hand x-ray; treat with topical calcipotriene (vitamin D derivative), topical steroid, or coal tar; if topicals not effective or patient has systemic psoriasis (arthritis), oral **methotrexate** or **acitretin** can be given; the latter is a vitamin A derivative.

- 8F + excoriations visible in flexor creases + occasional dry cough worse in winter; image of elbow shown below; Q asks what type of hypersensitivity is most likely responsible for this patient’s presentation:



- Answer = type I (immediate); diagnosis is atopy; image shows eczema (atopic dermatitis); patient has cough-variant asthma (1/3 of asthma patients only have dry cough, usually worse in the winter or with exercise); treat with oil-based emollient and topical corticosteroids; if steroids used >5-7 days continuously, thinning of the dermis may occur.

- 18F + history of eczema + area over elbow is red, inflamed, and oozing; 6-year-old sister recently had weeping papules on face; Dx in the 18-year-old? → answer = *Staph aureus* or Group A Strep superinfection over eczema; likely inoculated from younger sister's impetigo.
- 22M + recently treated with azithromycin for chlamydial urethritis; forearm is shown in below photo; Dx?



- Answer = urticaria (hives); type I hypersensitivity; can be precipitated by various allergens, including pollen, pet dander, and drugs.

- 39M + went hiking + used sunscreen over body + has linear vesicles on legs; Q asks best way to prevent this condition; Dx? → answer = "avoidance of contact with weeds"; diagnosis is contact dermatitis due to poison ivy/sumac; **linear vesicles** is hugely HY descriptor for poison ivy/oak/sumac; sunscreen will be the answer if they say rash on dorsa of hands, arms, and face (i.e., it's everywhere); nickel will be the answer for contact dermatitis if they mention vesicles on the wrist in someone who

wears a watch; USMLE also loves latex allergy as red rash on hands of someone using gloves as anatomy proctor or working as nurse in new facility, as well as adhesives allergy (i.e., from bandages in hospital); contact dermatitis is type IV hypersensitivity (T cell-mediated); rash will appear within days of exposure to irritant + **will take a few days to go away** following removal of the irritant.

- 38F + recently treated with trimethoprim/sulfamethoxazole for simple UTI + develops sloughing skin over her arms; image shown below; Dx?



  - Answer = Stevens-Johnson syndrome → autoimmune type IV hypersensitivity (T cell response) resulting in detachment of skin covering <10% surface area of body; Nikolsky sign is (+), which is sloughing of the skin with friction. Toxic epidermal necrolysis is sloughing of >30% surface area; 10-30% is an intermediate form; classically caused by sulfa drugs and anti-epileptics (lamotrigine).

- 24F + tattoo a couple days ago + image shown below; Q asks for Dx:



  - Answer = contact dermatitis from tattoo ink; asked on USMLE.

- 24F + nurse + allergic to bananas + image of hands shown below; Q asks diagnosis:



- Answer = Latex allergy (contact dermatitis; type IV hypersensitivity); patients who are allergic to bananas are often allergic to latex; contact dermatitis secondary to medical adhesives (i.e., bandages) can also occur, causing well-demarcated rash.

- 38F + chronic dry skin on the legs + scratches same area repeatedly; no other past medical history; image shown below; Dx?



- Answer = lichen simplex chronicus; dry, excoriated skin due to repeated scratching; dry, excoriated skin due to repeated scratching. The word excoriated means scratched. It's an important vocab word for Derm.

- 23M + painful vesicular lesions on the lip + fever + lymphadenopathy + presents with rash on arms as shown in below photo; diagnosis?



  - Answer = Erythema multiforme; immune complex-mediated rash (type III hypersensitivity) with many etiologies; HSV1/2 is classic cause; can also be caused by drugs (e.g., sulfa).

- 21M + blistering skin lesions + oral involvement + no mention of medications recently taken + Nikolsky sign (+); Q asks for the molecular target of the antibodies seen in this condition? → answer = desmosomes (desmoglein proteins), which mediate adjacent keratinocyte adhesion; diagnosis is pemphigus vulgaris, which is a blistering autoimmune skin condition caused by antibodies against desmosomes; oral involvement is common; Nikolsky sign is positive; immunofluorescence will show a net-like pattern.
- 21M + blistering skin lesions + no oral involvement + no mention of medications recently taken + Nikolsky sign negative; Q asks for the molecular target of the antibodies seen in this condition? → answer = hemidesmosomes, which stabilize basal epithelial cells to the basement membrane; diagnosis is bullous pemphigoid; less severe than pemphigus vulgaris; usually no oral involvement; Nikolsky is negative; immunofluorescence shows a linear pattern.
- 21M + formation of skin blisters with minor trauma; Q asks for what type of cell-cell interaction is disrupted in this patient? → answer = "basal:suprabasal" → Weird answer, but on NBME they write it just like that with the colon separating the words; diagnosis is epidermolysis bullosa, which is due to mutations (not antibodies, as with the aforementioned PV and BP) in keratin 5 and 14 of the dermal-epidermal junction; formation of blisters with trauma is not Nikolsky sign; the latter is removal / sloughing of the skin with friction, not blister formation with friction.
- 26M + oral + genital ulcers + high ESR; Dx? → answer = Behcet disease.
- 63M + patchy facial erythema that worsens with spicy foods and alcohol; slight pain of rash in cold weather; image of patient is below; Q asks simply for diagnosis:



- Answer = rosacea; multifactorial etiology; described as mix of itchy and painful facial rash in middle age patient that gets worse with things like cold weather, alcohol, and spicy food; topical metronidazole sometimes effective, as with avoidance of certain triggers like spicy food or alcohol.
- Rhinophyma is a severe bulbous enlargement of the nose that can occur in some patients with rosacea.

- 46M + high serum calcium + CXR shows bilar lymphadenopathy + image of face is shown below; Dx?



- Answer = lupus pernio (a cutaneous manifestation of ***sarcoidosis,*** not SLE, despite the name).

- 26F + presents during winter + painful/itchy toes + topical antifungals not effective; Dx?



- Answer = perniosis (chilblains); painful inflammation of distal capillaries due to repeated exposure to cold air, followed by immersion in hot water (i.e., from bath/shower).
- This is different from frostnip and frostbite. Frostnip is cold-exposed skin (effects quickly reversible; no skin damage); frostbite is more severe and can result in damage such as blistering and necrosis.

  - Can sometimes be intensely itchy and hence confused with tinea pedis.

- 28F + occasional painful single mouth ulcers (image shown below); Dx + Tx?



  - Dx = aphthous ulcers (aphthous stomatitis); no Tx necessary; **not** related to HSV1/2; etiology is multifactorial; can be precipitated by allergens such as spice; T cell-mediated.

- 35F + mouth ulcers + fever + treated two days ago for hyperthyroidism in hospital; Dx? → answer = Drug-induced neutropenia. Propylthiouracil and methimazole are thionamides (anti-thyroid drugs) that can both cause neutropenia (aka agranulocytosis), which can present as mouth ulcers (mucositis). Propylthiouracil is one of the drugs that is classically used for thyroid storm, so a recent hospitalization for hyperthyroidism suggests the patient was probably given this drug and now has neutropenia. HY drugs for neutropenia are the thionamides, clozapine, methotrexate, and TMP/SMX.

- 37M + occasional itchy bumps on hands; sometimes in ring-like pattern; Dx?



  - Dx = granuloma annulare; caused by cutaneous T cell response; occurs in young, healthy patients; may present as small bumps that progress to annular/ring-like pattern; self-resolving + no Tx necessary.

- 45F + hepatitis C positive + purple, pruritic skin lesions; image shown below; Dx?



- Answer = lichen planus; classically described as “The Ps” → purple, pruritic, polygonal papules; however Qs need not mention they’re pruritic; hepatitis C + red/purple skin lesions = lichen planus. I've also seen it occur idiopathically on NBME. USMLE wants you to know this often just self-resolves if no treatment is initiated.

- 28F + thrombocytopenia + polyarthritis + chronic sores of the cheeks and scalp + anti-Smith antibody positive; diagnosis? → answer = SLE; dermatologic component is discoid lupus, which is a severe cutaneous presentation seen sometimes in SLE patients, with sores/scarring of the face, scalp, and ears; thrombocytopenia (and leuko-/erythropenia) is seen in many SLE patients due to anti-hematologic cell line antibodies; arthritis is the most common presenting feature in SLE.
- Neonate born to mother who took methimazole during first trimester; photo of child is shown; what’s the diagnosis?



- Answer = aplasia cutis congenita; absence of skin on an area of scalp; can be caused by teratogens such as methimazole.

- 5M + second episode of edema of face, hands, and arms; patient is prescribed danazol; image of patient is shown below; what is the most likely mechanism for this condition?



- Answer = deficiency of C1 esterase *inhibitor* (not C1 esterase alone); diagnosis is hereditary angioedema; condition characterized by recurrent swelling of various bodily regions; danazol (androgen receptor partial agonist) causes liver to produce more C1 esterase inhibitor.

- 79M + purpura on forearms and dorsa of hands + normal hematologic studies; Q asks most likely cause for this condition; answer = normal age-related changes; diagnosis is senile purpura; increased fragility of dermal collagen and blood vessels secondary to normal senescence.
- 65M + worked in construction; yellow, thickened, coarsely wrinkled skin of forearms; Q asks most likely cause for this condition; answer = sun exposure; diagnosis is solar elastosis (UV light exposure).
- NBME Q asks best way to prevent sun damage; "avoidance of sun" not listed as answer (would be correct if listed); Q gives answers such as SPF 15, SPF 30, etc.; correct answer = "wear protective clothing"; sounds obvious, but I've seen numerous students choose SPF 30, thinking there's a trick.
- 33F + third trimester of pregnancy + has itchy erythematous/violaceous rash on abdomen within stretch marks; image is shown below; Dx + Tx?



- Diagnosis = pruritic urticarial papules and plaques of pregnancy (PUPPP); "weird diagnosis," but asked on 2CK for obgyn; cause is sporadic/multifactorial; classically presents as itchy papules within **stretch marks;** oral antihistamines (2nd gen H1 blocker) are Tx.

- 33F + third trimester of pregnancy + itchy rash around umbilicus + stretch marks not involved; image is shown below; Dx + Tx?



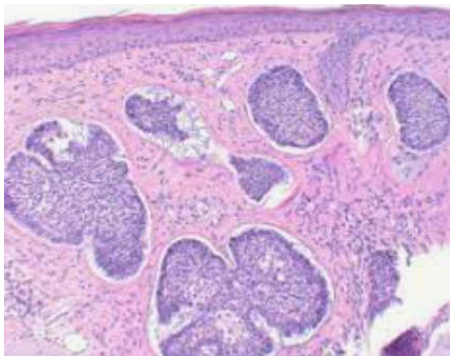
  - Dx is gestational pemphigoid; does not involve stretchmarks (unlike PUPPP); cause is sporadic/multifactorial; Tx is oral or topical steroids.
- 33F + third trimester of pregnancy + intense, diffuse pruritis, especially on palms + soles; no skin rash; serum bile acids are elevated; Dx + Tx? → answer = intrahepatic cholestasis of pregnancy; Dx by measuring increased serum bile acids; Tx is ursodeoxycholic acid (ursodiol).
- 12M + nosebleeds for the past week + petechial rash + bleeding time 9 minutes + platelet count 90,000/uL; Dx + Tx? → answer = idiopathic (immune) thrombocytopenic purpura (ITP); cause is antibodies against GpIIb/IIIa on platelets (type II HS); Dx with decreased platelet count (answer on NBME; increased bleeding time for Dx is wrong answer). Tx = steroids first, then IVIG, then splenectomy is definitive.
- 67M + lesion shown on nose in photo below; Q simply asks the diagnosis:



  - Answer = basal cell carcinoma (BCC); on USMLE, classically pearlescent / slightly translucent; talengiectasias common; can be described as having “heaped up borders” or “rolled edges”; may or may not be ulcerated.

- Treatment for skin cancers on cosmetically sensitive areas such as the eyelid or nose can be managed with Mohs micrographic surgery (on NBME).

- 30F + atypical skin lesion on neck + biopsy shows clusters of basophilic cells (shown below); Q asks for the diagnosis:



- Answer = BCC; histo shows clusters of basophilic (purple) cells.

- 42M + receives topical immunomodulator for confirmed BCC; what is the drug he received? → answer = imiquimod; stimulates toll-like receptor-7 (TLR-7).

- 62F + farmer + lesions on forearm/hand shown below; Q asks diagnosis:



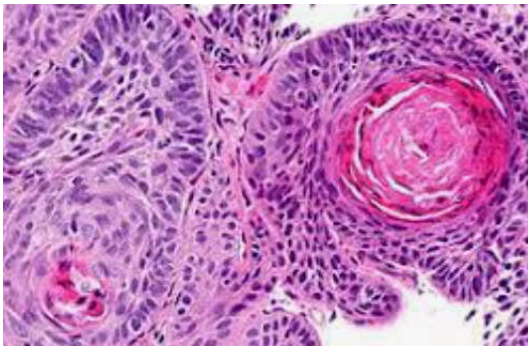
- Answer = actinic keratoses (aka solar keratoses); precursor to squamous cell carcinoma (SCC); classically described as red/scaly lesions on forehead, ear, or arms of fisherman, farmers, or construction workers. Can also appear as rough whiteish lesions on the hands/forearms. Cryotherapy is an answer for treatment on NBME.

- 74M + fisherman; forehead and ear are shown below; Q asks for the diagnosis of the ear lesion:



  - Forehead shows actinic keratoses. Ear shows squamous cell carcinoma; **actinic keratoses can classically progress to SCC;** patient has many actinic keratoses and was likely at risk of developing SCC, as with the ear lesion; on USMLE, SCC will not have telangiectasias or pearlescent/translucent appearance (as with BCC).
  - Both SCC and BCC can be ulcerated and/or have rolled edges (I've seen rolled/ulcerated SCC on NBME exam, even though rolled edges are textbook BCC. But with that particular Q the patient had actinic keratoses, so you know SCC is what progresses from that).
  - Do not memorize BCC vs SCC as necessarily occurring in certain locations (i.e., upper vs lower lip, etc.). USMLE has zero regard for this stuff. You need to look for telangiectasias and pearly appearance for BCC, versus absence for SCC.

- 60M + 2 packs cigarettes daily for 30 years + lesion on back; biopsy is shown below; Q asks Dx:



  - Answer = SCC; histo shows keratin pearls (pink circles), which are classic for SCC; USMLE wants you to know "keratin pearls + intercellular bridges" = SCC.
  - **Smoking + immunodeficiency** (e.g., HIV) are HY risk factors for SCC.

- 70F + chickenpox scar on chin since childhood + recent abnormal growth of the scar + biopsy confirms neoplasia; Q asks for the diagnosis → answer = SCC → **Marjolin ulcer** is the name given to an SCC

that arises from a prior scar or site of trauma/burn. This is an important factoid Dx for 2CK surgery as well.

- 68F + rough-surfaced grey/white lesion on labia majora; Dx + Tx? → answer = lichen sclerosus; described as rough, white/grey, patchy, and/or itchy area on vulva or perineum; must do biopsy to rule out SCC *before* topical steroids; the latter are effective treatment. If the Q gives an overtly ulcerated lesion on the vulva in this demographic, the diagnosis is vulvar carcinoma, where lichen sclerosus is the wrong answer. LS will not be described as ulcerated.
- 65F + lesion on face shown below; Q asks for the diagnosis:



    - Answer = keratoacanthoma; can be confused with SCC; described as fast-growing and dome-shaped with a hyperkeratotic core, surrounded by a wall of inflamed skin; can sometimes self-regress, but Tx is still surgical excision.
- 23M + 1-cm lesion on arm as shown in photograph below + lesion has not changed recently + uncle died of melanoma; what's the next best step in management?



  - Answer = observation; diagnosis is nevocytic nevus (benign mole); when choosing excisional answers on USMLE, think ABCDE → Asymmetry, Border (irregular), Color (variegated), Diameter (>1cm), Elevation.

- 45F + lesion on leg shown in photo below; what's the next best step in management?



  - Answer on NBME = excisional biopsy; suspected malignant lesions in non-cosmetically sensitive areas (i.e., not on the head/neck) can simply be excised with narrow margins; excision of additional tissue is important if margins are positive (i.e., entire lesion wasn't excised).
  - "Full-thickness biopsy" answer on USMLE for suspected melanoma on back of neck (NBME Q gives lesion similar to above on neck, with full-thickness biopsy as answer). Excision of lesion is management for neck lesions if full-thickness biopsy confirms melanoma; do Mohs micrographic surgery for facial lesions.
  - Punch biopsy is a type of full-thickness biopsy; if Q asks, choose side of lesion for biopsy, rather than middle of lesion.
  - Students will sometimes ask about shave biopsy. I have never seen this as correct answer on NBME, but literature says it can sometimes be used to remove superficial non-pigmented lesions where the clinician does not suspect melanoma (performing shave biopsy on melanoma can create problems for assessing depth, prognosis, and therapy).

- 45M + confirmed melanoma; Q asks which aspect most relates to prognosis; answer = depth of lesion. Wrong answer is "lymphocytic infiltrate"; the latter is *good* for prognosis, as immunosurveillance functions to suppress skin cancers. Depth of lesion is most important for prognosis.

- 32F + confirmed melanoma + receives aldesleukin; Q asks which cytokine this relates to? → answer = IL-2; aldesleukin is a recombinant interleukin-2 that can be used in the treatment of melanoma and RCC; IL-2 normally functions to stimulate T cells.
- 30F + African-American + lesion on foot as shown below; what's the diagnosis?



  - Answer = acral lentiginous melanoma; melanoma of palms/soles (areas not usually exposed to sun); more common in persons of African and Asian descent.
- 34F + Asian-American + fingernail shown in photo below; diagnosis?



  - Answer = acral lentiginous melanoma; USMLE will show you image of ALM on the palm/sole, or it can be subungual (as shown here).
- 29M + nailbed lesion + radiating pain incited by cold temperature; Dx? → answer = glomus tumor; The glomus body is a type of modified smooth muscle cell in fingers/toes that assists in thermoregulation. Sounds arcane/weird, but for some magical reason exists on USMLE.
- 67M + lesion on face as shown in photo below; biopsy shows malignant cells growing laterally along the basement membrane without dermal invasion; Q asks diagnosis:



- Answer = lentigo maligña; considered melanoma *in situ*; starts as black/brown "stain" that grows laterally within stratum basale without penetration; once it invades, it is called lentigo maligña melanoma (Hutchinson melanotic freckle); the terminology can sound confusing because lentigo maligña is still technically melanoma, albeit *in situ*, but once it invades it takes on "melanoma" as the suffix. This is on NBME exam.

- 37F + confirmed melanoma + two other first-degree family members also had melanoma; Q asks for which gene is most likely associated in this patient → answer = *BRAF*; proto-oncogene, not a tumor suppressor; *BRAF* codes for BRAF serine-threonine kinase. You do not need to know vemurafenib (targets BRAF kinase) for USMLE, but some students have asked about it.
- 83F + facial lesions shown in photo below; Q asks for diagnosis:



  - Answer = seborrheic keratoses; sun exposure and old age are biggest risk factors; described as "greasy," or waxy, skin growths that appear like they can be "peeled off"; not malignant.
  - Sign of Leser-Trélat = sudden, eruptive seborrheic keratoses secondary to underlying, visceral malignancy (i.e., unrelated to age or sun).
- 23M + skin-colored, painless 4-mm growth on penile shaft; Q asks most likely viral etiology; answer = human papillomavirus 6 or 11; cause condylomata acuminata (warts); HPV 6 and 11 can also cause laryngeal papillomatosis (vocal cord growths) in pediatrics (due to vertical exposure from birth canal); HPV 16 and 18 are HY strains for cervical/vaginal/penile SCC.
- 55M + fungating mass from the rectum; Q asks next best step in Dx? → answer = biopsy of the mass (simple Q on surgery form); likely SCC from HPV 16/18.
- 55M + 3-month Hx of pencil-like stools + 3-cm fungating mass just inside anal verge; Q asks next best step in management (biopsy of mass not listed) → answer = colonoscopy; wrong answer is "surgical

excision of mass" and chemo/radiotherapies. Apparently colonoscopy first needs to be done to evaluate for extent of colonic involvement before definitive surgical management is employed.

- 15M + mouth shown in photo below; Q asks for what kind of polyps he most likely has:



  - Answer = hamartomatous; Dx is Peutz-Jeghers syndrome; combination of perioral melanosis (sophisticated way of saying hyperpigmentation around the mouth/lips) and hamartomatous colonic polyps.
  - Perioral melanosis also seen in Carney complex (cardiac myxoma *in a kid*, perioral melanosis, endocrine hypersecretion). Not fucking with you. It's on the USMLE.
- 6M + jaw lesion as shown in photo below; Q asks for molecular function of gene involved:



  - Answer = transcription factor; Dx is Burkitt lymphoma; usually t(8;14) translocation; *c-MYC* gene; codes for transcription factor; "starry sky" appearance on histo.
  - Focus here isn't heme/onc, but HY contrasting point: do not confuse with follicular lymphoma, which is t(14;18) translocation, *BCL-2* gene; codes for anti-apoptotic molecule; presents as waxing/waning neck mass over 1-2 years.
- 49M + smoker + rough, white lesion on lateral aspect of tongue that does not scrape off; Dx? → answer = leukoplakia; precancerous lesion to SCC; caused by smoking / chewing tobacco.
- 32M + HIV positive + white lesions on lateral tongue that do not scrape off; Q asks for the viral etiology → answer = EBV; Dx is oral hairy leukoplakia; not precancerous; caused by hyperkeratosis

(excessive skin proliferation); one of the most common presentations of HIV (in addition to oropharyngeal candidiasis).

- 54M + IV drug-user + lesion on leg shown in photo below + biopsy shows malignant T cells with cerebriform nuclei; Q asks which virus is most likely responsible:



  - Answer = Human T cell lymphotropic virus (HTLV-1/2); diagnosis is mycosis fungoides, which is a cutaneous T cell lymphoma; T cells have nuclei with classic “cerebriform” appearance; Pautrier microabscesses are atypical T cells in epidermis; HTLV-1/2 is retrovirus (RNA, enveloped) most similar to HIV; increased prevalence in IV drug-users and persons in Japan and Caribbean; you do not need to be able to do a spot-diagnosis here; Q will pretty much always give you enough descriptors in the vignette.

- 54M + Hx of IV drug-use + diffuse red body rash as shown in photo below + blood smear shows leukemic T cells with cerebriform nuclei; Q asks for structure of viral etiology (i.e., RNA vs DNA; enveloped vs non-enveloped):



  - Answer = RNA, enveloped; virus is HTLV-1/2; diagnosis is Sezary syndrome; T cell leukemic extension of mycosis fungoides; diffuse exfoliative erythroderma is classic.

- 82F + had gallbladder removed at age 58 + afebrile + abdomen shown below; Q asks next best step in diagnosis?



  - Answer = CT of abdomen with contrast; Dx is pancreatic cancer; image shows jaundice (yes, I jacked up the saturation to make the patient yellow AF); remote cholecystectomy denotes impossibility of choledocholithiasis; this Q asked on Surg NBME; if the Q tells you in the last line that CT shows no abnormalities, next best step = ERCP to look for cholangiocarcinoma (bile duct cancer).
  - Head of pancreas cancer impinges on common bile duct → jaundice with increased ALP and direct bilirubin; pancreatic enzymes are *normal*; if gallstone pancreatitis, enzymes up.
- 40M + history of shooting groin pain + elevated serum glucose + red rash on abdomen; Q asks for which pancreatic cancer type is the Dx → answer = glucagonoma → can cause ↑ glucose + necrolytic migratory erythema (rash). Patient has MEN1 (pancreas, parathyroid, pituitary); shooting groin pain = ureterolithiasis; hyperparathyroidism → high Ca^{2+} → stones.
- 40M + flushing of face + diarrhea + low serum potassium; Dx? → answer = VIPoma; classically WDHA syndrome → Watery Diarrhea, Hypokalemia, Achlorhydria; can also cause facial flushing.
- 27F + 20 weeks' gestation + scattered skin lesions on trunk and arms, as shown in photo below; patient also has axillary/groin freckling; Q asks inheritance pattern:



- Answer = autosomal dominant; Dx is neurofibromatosis type I (NF1); image shows café au lait spot; one of the phakomatoses (neurocutaneous disorders → NF1/2, VHL, TSC, Sturge-Weber); NF1 → café au lait spots, neurofibromas, axillary/groin freckling, optic glioma, pheochromocytoma; for some reason, USMLE likes NF1 in obgyn Qs, even though the condition is genetic / unrelated to obgyn.

- 7M + lesions shown on face in image below; Q asks inheritance pattern:



  - Answer = autosomal dominant; Dx is tuberous sclerosis (TSC); image shows adenoma sebaceum (angiofibromas); one of the phakomatoses; in TSC we see intracranial / periventricular nodules (tubers, aka cortical hamartomas), adenoma sebaceum, cardiac rhabdomyoma, lymphangeoleiomyomatosis, subungual fibromas, renal angiomyolipoma, Ashleaf spots, and Shagreen patches.

- 17M + Hx of epilepsy + physical examination shows violaceous papules in a temporal distribution; Dx? → answer = Sturge-Weber; classically Port Wine-stain birthmark (nevus flammeus) in textbooks, but USMLE can describe this as “violaceous papules in a temporal distribution”; associated with leptomeningeal angioma (causing seizure) and glaucoma.



- Classic Port wine stain birthmark, as seen in Sturge-Weber; condition is not inherited and is due to somatic mosaicism.

- 3M + deficiency of α-galactosidase A + buildup of ceramide trihexoside + heart/kidney issues + red lesions on skin shown in image below; Q asks diagnosis:



  - Answer = Fabry disease; lysosomal storage disease; X-linked recessive; image shows angiokeratomas, which are dilated capillaries forming papules.

- 6M + skin lesions as shown in image below + polyostotic fibrous dysplasia + testes large for gestational age; Dx?



  - Answer = McCune-Albright syndrome → triad of "coast of Maine" café au lait spots (above image), polyostotic fibrous dysplasia (bone is replaced by fibrous tissue), and endocrine hypersecretion (classically precocious puberty).

- 3M + retained primary teeth + eczema + recurrent *Staphylococcal* abscesses; Dx? → answer = hyper-IgE syndrome (Job syndrome) → FATED → coarse Facies, *Staphylococcal* Abscesses, retained primary Teeth, hyper-IgE, Dermatologic abnormalities (eczema).
- 3M + delayed separation of umbilical cord at birth + recurrent skin infections without pus + biopsy shows decreased neutrophils at sites of skin infection; Dx? → answer = leukocyte adhesion deficiency; defective LFA-1/CD18 integrin.
- Neonatal male + family recently immigrated to US from China; back is shown in image below; Dx?



  - Diagnosis is Mongolian spot (blue nevus), a form of benign birthmark where dermal melanocytes fail to migrate superficially to stratum basale. Often mistaken for child abuse. Schedule routine follow-up. Wrong answer is contacting child protective services.
- 22M + BMI 36 + neck shown in image below; what is most likely to be seen in this patient?



  - Answer = hyperinsulinemia; image is acanthosis nigricans (brown/black velvety skin, usually along nape of neck); associated with hyperinsulinemia and type II diabetes; can also rarely be seen in patients with underlying, visceral malignancy (e.g., gastric adenocarcinoma).

- 42M + lantern jaw + increased hat/shoe size + high BP + image shown below; Q asks next best step in diagnosis:



  - Answer = measure serum insulin-like growth factor I (IGF-1); wrong answer is measure serum growth hormone; diagnosis is acromegaly; GH causes liver to secrete IGF-1, which in turn promotes growth of tissues; skin tags can be seen secondary to insulin resistance; growth hormone excess causes insulin resistance (tangential, but for Step 1, choose ↑ activity for catabolic/gluconeogenic enzymes; ↓ activity for anabolic enzymes, since ↓ insulin effect).
- 3M + coloboma of the iris + ventricular septal defect + ear is shown below; Q asks what else is likely to be seen in this patient?



  - Answer = atresia of the choanae; Dx is CHARGE syndrome → Coloboma of the iris, Heart defects, Atresia of the choanae (baby turns blue during breastfeeding and pink when crying; Dx by inserting nasogastric tube), Retardation of growth/development, Genitourinary anomalies, Ear anomalies (protruding ears with lack of earlobe).

- 7M + hole seen on preauricular aspect of ear bilaterally; patient is asymptomatic; Q asks treatment:



  - Answer = no Tx necessary; preauricular pits are benign finding, usually not associated with any congenital disorder; can become infected; surgical closure if frequent infections.

- Neonate + lesion superior to pinna in image shown below; Dx+ Tx?



  - Dx is strawberry hemangioma; benign capillary tumor in neonates that will grow slightly then regress spontaneously within a few years; no Tx necessary unless causing functional impairment.

- 50F + photo of lesions on neck shown below; Dx + Tx?



  - Answer = cherry hemangiomas; no Tx necessary; benign capillary tumors caused by UV light; greater prevalence with increasing age. Students confuse with strawberry hemangioma in peds.

- Neonate + tuft of hair seen on lower back; Dx? → answer = spina bifida occulta; NBME exam has one answer written as "spinal dysraphism," rather than spina bifida.
- 50M + long-standing Hx of ulcerative colitis managed with multiple medications; abdomen shown in image below. What is causing this skin presentation:



  - Cushing syndrome due to exogenous prednisone used in autoimmune disease; USMLE wants you to know purple striae are classically Cushing. This is due to weakening of dermal collagen and capillary walls, with micro bleeding into skin.
  - Cushing syndrome can also cause hyperpigmentation due to ↑ ACTH in some patients, including patients with small cell who have ectopic ACTH secretion.

- 41M + serum glucose 130 mg/dL + sore hands + hand x-ray shows Heberden nodes; physical exam shows darkening of the skin of the forearms; Q asks next best step in diagnosis? → answer = check serum ferritin; Dx is hereditary hemochromatosis; can present as "bronze diabetes" → diabetes due to deposition of iron in tail of pancreas; hemosiderin deposition in skin causes hyperpigmentation (hemosiderosis); arthritis is pseudogout, not osteoarthritis (two of the biggest risk factors for pseudogout are hereditary hemochromatosis and primary hyperparathyroidism; can present as OA-like presentation in someone with aforementioned conditions, or as monoarthritis of large joint, such as the knee).
- 30M + 6-month-Hx of fatigue + serum potassium 6.0 mEq/L, sodium 130 mEq/L, bicarb 20 mEq/L + eosinophils 23% + hyperpigmentation of forearms; Q asks which diagnostic test should be performed? → answer = ACTH stimulation test; Dx is Addison disease (primary hypoadrenalism); classically low

sodium, high potassium, and low bicarb when aldosterone is low; low cortisol can cause eosinophilia; do not go chasing stool ova and parasites; hyperpigmentation in the setting of low cortisol is due to decreased negative feedback at the hypothalamus and anterior pituitary → increased POMC, which is the precursor to both ACTH and α-MSH; both ACTH and α-MSH can increase pigmentation.

- Neonate + violaceous lesion on leg shown in photo below; platelet count 50,000/uL; Q asks mechanism for thrombocytopenia:



  - Answer = platelet sequestration; Dx is Kasabach-Merritt syndrome (aka infantile hemangioma with thrombocytopenia); asked several times on 2CK Peds assessment; not strawberry hemangioma; platelets can be sequestered within lesion; Tx is surgical.

- 3M + brought in by mother for pain in right arm after falling off swing on playground; x-ray shows fracture of the humerus; photo of hand shown below; Q asks next best step?



  - Answer = contact child protect services; photo shows classic circular appearance of cigarette burn; can also be on face / resemble impetigo if at different stages of healing; humerus fractures uncommon and can be from blunt-force trauma; child abuse classically spiral fractures (rotational force), but emphasis in above Q is the cigarette burn.

- 22F + lesions on ear shown in image below + occurred following removal of ear piercings several weeks ago; Dx?



  - Answer = keloid scars; disorganized growth of collagen type I and III; scar grows beyond boundaries of original wound; In contrast, hypertrophic scars may resemble keloids but do not grow beyond boundaries of original wound; Tx is surgical excision, although recurrence is common. Benign.

- 22M + face shown in photo below; vitals normal; no past medical Hx; Dx?



  - Answer = pseudofolliculitis barbae (razor bumps); increased prevalence in African descent; curly beard hair grows back into the skin; Tx is to allow the beard to grow; USMLE Q will show you image and just ask the spot-Dx.

- 65F + Hx of pain in buttocks/thighs when walking his dog + image of foot shown; what's the most likely cause of this finding?



- Answer = arterial insufficiency; image shows arterial ulcer; often punched-out in appearance, well-demarcated; usually tops/bottoms of feet/toes; patients will frequently have Hx of intermittent claudication, diabetes, CABG, etc. Peripheral pulses will be decreased (i.e., popliteal and dorsalis pedis). Arterial disease is often associated with trophic changes of legs demonstrating shiny skin with loss of hair.

- 50F + image shown below; what's the most likely cause of this finding?



- Answer = chronic venous insufficiency; image shows venous ulcer; classically large, sloughy lesion around the ankle / medial malleolus; hyperpigmentation common in venous disease due to congestion and hemosiderin extravasation ("brawny edema"). Peripheral pulses are normal in venous disease.
- USMLE wants you to be able to spot-diagnose arterial vs venous ulcers. Exceedingly HY for 2CK surgery Qs in particular.

- 54M + leg shown below; no other past medical Hx; Q asks for next best step in Dx + Tx?



- Image shows varicose veins; Dx → Duplex venous ultrasonography for venous disease / valvular insufficiency; Tx is compression stockings; do not choose surgical interventions for varicose veins on USMLE.

- 50M + 1-cm painful, palpable “cord” around the ankle / tracks up toward the knee; image is shown below; Q asks for Tx:



  - Answer on NBME = subcutaneous enoxaparin; Dx is superficial thrombophlebitis; similar to DVT, but in more superficial vein; treated with heparin; wrong answer is compression stockings; difficult Q since compression stockings common answer for venous disease, but if patient has active venous occlusion (i.e., DVT or superficial thrombophlebitis), give heparin.
- 48M + heavy smoker + hands shown below + no other significant past medical history; Dx + Tx?



  - Dx = thromboangiitis obliterans (Buerger disease); Tx = smoking cessation; presents as digital gangrene in males who are heavy smokers.
- 66F + type II diabetes + HbA1c 9.8% + metoclopramide is one of her meds; image of foot shown below; x-ray of the foot shows disorganization of the tarsals/metatarsals; Q asks most likely cause of the lesion shown:



- Answer = “lack of appropriate joint sensation”; Dx = Charcot joint (neuropathic joint); patient cannot feel her feet due to peripheral neuropathy; vignette description points away from pure arterial ulcer (plus arterial ulcers are more distal/punched-out, rather than around the ankle); metoclopramide implies patient has diabetic gastroparesis, hence is on the prokinetic; gastroparesis implies advanced neuropathy; USMLE can give you Hx of neurogenic/hypotonic bladder as well (i.e., on bethanechol).

- 62M + hepatitis C + decreased serum C4 + legs shown in image below; Dx?



- Answer = cryoglobulinemia (presenting as livedo reticularis) secondary to hepatitis C; cryoglobulins are immune complexes that precipitate at cold temperatures (type III HS); livedo reticularis is a mottled, reticulated vascular pattern that has many etiologies; cryoglobulinemia is associated with decreased serum complement protein C4 (in contrast, C3 is sometimes down in SLE flares and Group A Strep infections).

- 65M + smoker + appearance shown in photos below; Dx?



- Answer = Pancoast tumor causing SVC syndrome; photos show Pemberton sign (facial erythema/plethora due to diminished venous return when arms are raised above head).
- Children with ALL who have (+) Pemberton sign have T cell variant (thymic lesion); normally ALL is B cell, without thymic lesion.

- 49M + alcoholism + image shown below; Q asks which vessel(s) is/are experiencing congestion:



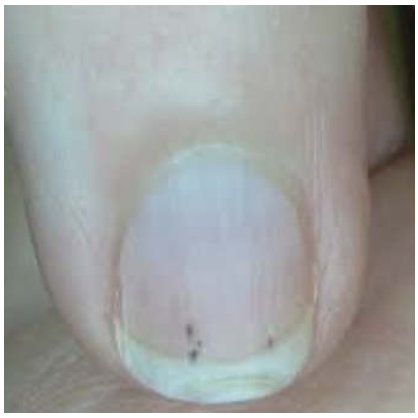
- Answer = superficial epigastric veins; Dx is caput medusae; sometimes seen in severe portal hypertension secondary to cirrhosis.
- Don't confuse with "left gastric vein," which is often the answer for esophageal varices (i.e., portal → left gastric → esophageal veins).

- 70F + chronic alcoholism + hands shown below; Q asks mechanism for this finding:



- Answer = “failure of the liver to degrade estrogen”; palmar erythema (above image), spider angiomata, and gynecomastia are classic hyper-estrogenic findings seen in advanced liver disease; Q can also have “hyperestrogenism” as the answer; if the latter isn’t listed, “decreased testosterone” is also correct on one of the NBMEs.

- 38F + was told when she was younger she had a “click” in her heart + recent dental procedure + fever of 103 F + 3/6 holosystolic murmur; finger is shown below; Q asks next best step in management:



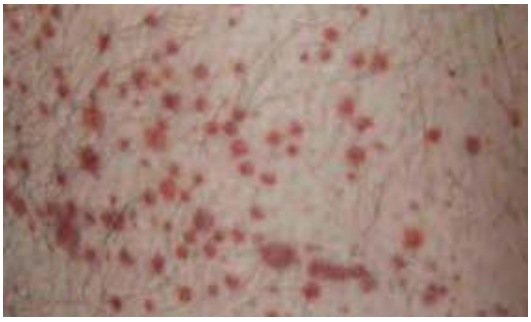
  - Answer = blood cultures; diagnosis is subacute endocarditis due to *Strep viridans* (Hx of dental procedures in someone with valve abnormality; this patient has Hx of mitral valve prolapse); do blood cultures *before* antibiotics; then do transesophageal echocardiogram (TEE) to diagnose; murmur above is mitral regurg (preceding MVP would have been mid-systolic click). Patients can also have Janeway lesions / Osler nodes in endocarditis.

- 70M + smoker + recently underwent AAA repair + foot shown below; Dx?



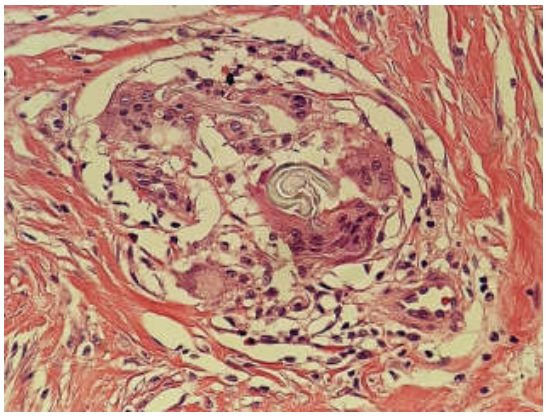
  - Answer = cholesterol emboli; due to embolization of cholesterol plaques to distal vasculature; Hx of AAA repair classic; atheromata launch off to distal arterioles/capillaries.

- 24M + snowboarding accident where he collided with tree + severely painful and distorted left thigh + platelets are 50,000/uL; on examination, patient’s chest is shown below; Dx?



- Answer = fat embolism; patient has femoral shaft fracture → release of bone marrow fat into blood → chelates platelets → thrombocytopenia; **petechial rash on chest** due to occlusion of microvasculature by fat; thrombocytopenia in and of itself is not responsible for petechial rash (for instance, this finding is not classic in ITP).

- 29M + laparotomy 3 months ago following gunshot wound + now has erythematous, tender nodule along incision line + afebrile; patient is treated appropriately and light microscopy of lesion is shown below; Dx?



- Answer = suture granuloma; immune reaction from residual foreign suture material; Tx is removal of suture. NBME Q shows histo revealing foreign body surrounded by histiocytes (activated macrophages).

- 49M + rash over axillae + alopecia + diminished smell and taste; Q asks which nutrient is deficient → answer = zinc; classically causes anosmia (decreased smell) + hypogeusia (decreased taste), but the vignette need not mention these findings. Can also cause alopecia and rash. Neonates with congenital malabsorption (acrodermatitis enteropathica) can have rash on face.

- 82F + “appears ill” + bruises on forearms + bleeding around hair follicles and from gums; Dx? → scurvy (vitamin C deficiency); perifollicular hemorrhages and oral mucosal bleeding are classic.

- 70M + image of neck shown below; Q asks for nutrient deficiency:



  - Answer = niacin (vitamin B3); image shows Casal necklace; B3 deficiency causes pellagra → 3Ds → dementia, dermatitis, diarrhea; the photodermatitis is classically Casal necklace, but can also be described as merely hyperpigmentation of the forearms in a patient with increased bowel motions and delirium (one of the biggest risk factors for delirium is underlying dementia).

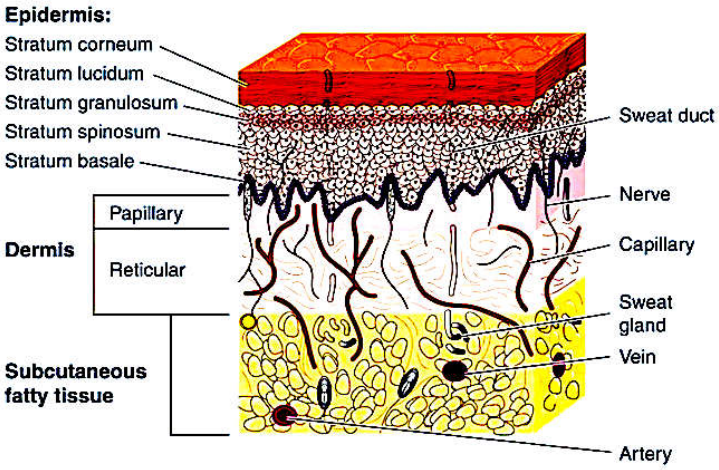
- 50M + drinks plenty of alcohol and bags of nucleic acids; image of painful toe is shown below; Tx?



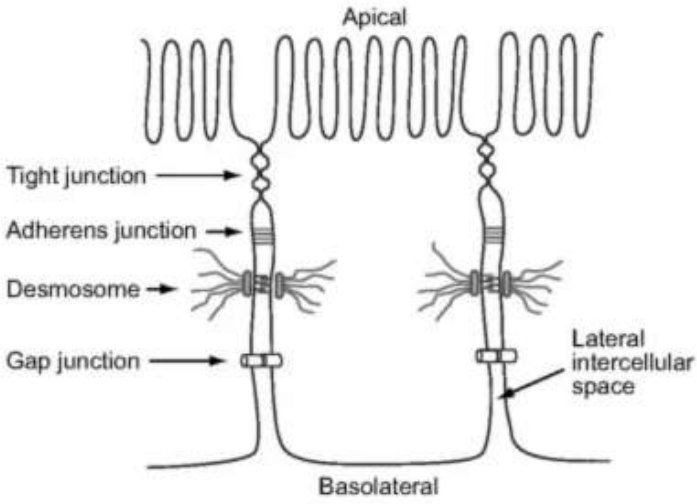
  - Answer = indomethacin (NSAID), colchicine (microtubule inhibitor), or prednisone; diagnosis is acute gout; chronic gout managed first-line with xanthine oxidase inhibitor (i.e., allopurinol or febuxostat) between attacks.

- "What are the layers of the skin?"

  - You might say, "Yo that's weird you didn't mention this first thing cuz it's hyper-basic." → Reason I held off is because USMLE doesn't directly assess, "What are the skin layers?" But it would be odd not mentioning them at all since this is literally a dermatology PDF.



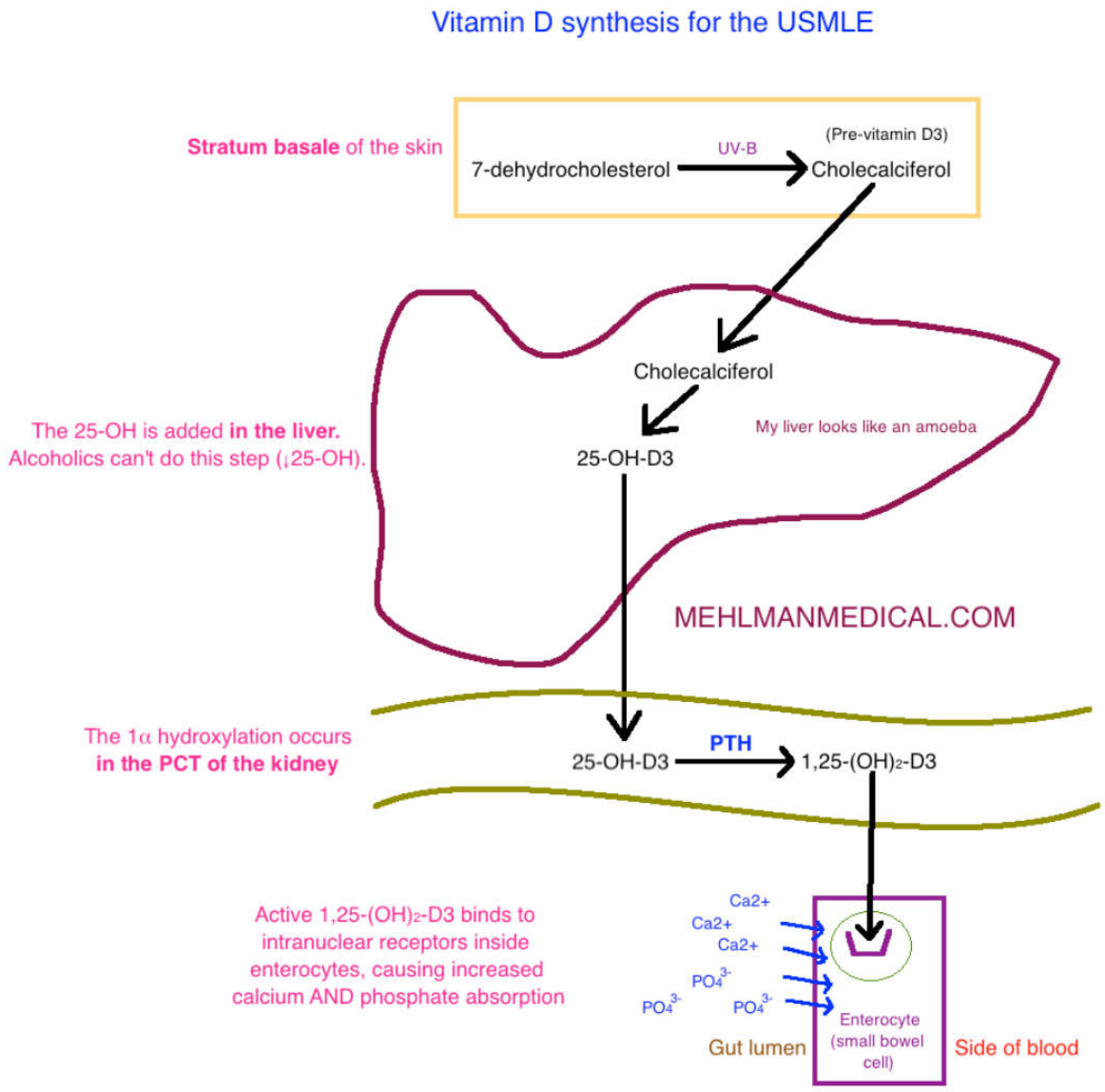
- 25F + long time spent in sun + erythema over 70% of her body; Dx + Tx? → answer = first-degree burn (sunburn); Tx = supportive care / no Tx necessary; do not select answers such as silver sulfadiazine, etc.
    - First-degree (superficial) burn → epidermis only → diffuse erythema + pain.
    - Second-degree (superficial partial) → involves papillary dermis → blistering + pain.
    - Second-degree (deep partial) → involves reticular dermis → white/leathery; painless.
    - Third-degree (deep; full-thickness) → involves hypodermis → painless.
    - Fourth-degree → involves underlying muscle, tendon, or bone.
    - I have not seen silver sulfadiazine and triple antibiotic ointment as answers on NBME assessments; literature is mixed on their use.
- 20M + body weight 75kg + 30% of body surface area third-degree burned in housefire; 2CK Surg Q asks how much / what type of fluid should be given:
    - Parkland formula for surgery is used to calculate fluid resuscitation over next 24 hours.
    - Parkland formula = 4mL x (% surface body area) x (weight in kg).
    - 4 x 30 x 75 = 9,000 mL = 9L Ringer Lactate or 0.9% normal saline over next 24 hours.
    - Give first half in first 8 hours post-burn; give second half in subsequent 16 hours.
    - Yes, there is a calculation Q on Surg NBME for 2CK.
- “What do I need to know about dumb molecular skin stuff? Like the connective proteins and stuff.”



- Tight junction is the answer if they ask what prevents movement of solute between cells (i.e., prevents water from getting through your skin). They can also ask about this in relation to gastrointestinal mucosa (i.e., person with IBD demonstrates radiolabeled substance in blood after it was injected into bowel; however person without IBD does not have the substance in the blood; why? → answer = loss of tight junction function in IBD patient. Tight junctions contain proteins called claudins and occludins.
- Adherens junction is the answer if they ask about what connects the actin cytoskeleton of adjacent cells. E-cadherins are essential proteins (E-c**adherins** compose **adherens** junctions); E-**c**adherins are **c**alcium-dependent.
- Desmosome is the answer for pemphigus vulgaris and staphylococcal scalded skin syndrome; composed to desmoglein proteins.
- Gap junctions contain connexin proteins. Unrelated to skin, but I've seen this asked for myocardial muscle electrical conduction (i.e., how is electrical synchrony accomplished?) → answer = gap junctions. I've seen "connexin" as distractor answer on NBME, but never as correct answer.

- USMLE Q asks which layer vitamin D synthesis starts in → answer = stratum basale.
  - 7-dehydrocholesterol in stratum basale, via UV-B radiation, → cholecalciferol.
    - NBME Q tells you patient does not get sunlight and asks "synthesis of which substrate is impaired?" → answer = cholecalciferol; 7-dehydrocholesterol is wrong answer.

- Cholecalciferol goes to liver and takes on 25-hydroxylation, becoming 25-OH-D3 (calcidiol).
  - If patient has liver disease, 25-OH-D3 is ↓; answer can also be "decreased hepatic hydroxylation."
- 25-OH-D3 goes to the kidney and takes on the 1-hydroxylation, via 1α-hydroxylase, under the action of parathyroid hormone → 1,25-$(OH)_2$-D3 (calcitriol).
- 1,25-$(OH)_2$-D3 then goes to the small bowel and increases absorption of Ca^{2+} and $PO4^{3-}$; it also goes to bone and converts unmineralized osteoid → mineralized hydroxyapatite.



- "Do I need to know stuff like macule, patch, plaque, etc.? Various terminology?" → Yes, but this stuff more just applies to terms thrown around in USMLE vignettes. In other words, you will not get asked directly, "Is this a patch?" Or, "Is this a vesicle vs bulla?" Etc.

- **Macule:** flat lesion <1cm.
- **Patch:** flat lesion >1cm; classically café au lait spots in NF1.
- **Papule:** raised lesion <1cm; "maculopapular" → many viral rashes and drug reactions.
- **Plaque:** raised lesion >1cm; psoriasis.
- **Vesicle:** clear-fluid collection <5mm; HSV1/2, or shingles (VZV).
- **Bulla:** clear-fluid collection >5mm; bullous impetigo (*S. aureus*); pemphigus vulgaris, bullous pemphigoid, epidermolysis bullosa.

- 26F + brought in by ambulance following motorcycle accident + BP 160/100 + HR 52 + respiratory rate 10; image of patient shown below; Q asks mechanism for patient's hypertension:



  - Answer = "increased intracranial pressure"; image shows Battle sign (bruising over mastoid process seen in base of skull fracture); patient can also have racoon eyes, rhinorrhea, and otorrhea in base of skull fracture; Cushing reflex = hypertension, bradycardia, and bradypnea as a result of increased intracranial pressure.

- 25M + 2nd degree burn on leg from playing with fireworks + heals; 3 months later, area of burn appears darker than surrounding skin; Dx + Tx? → answer = post-inflammatory hyperpigmentation; benign hyperpigmentation that can occur following inflammation (as the name implies); no Tx necessary; usually self-resolving; if topicals used, hydroquinone or tretinoin often effective.
- 32F + on combined oral contraceptive pills for 12 years + slightly hyperpigmented skin on cheeks and buttocks; Dx + Tx? → answer = melasma (chloasma); benign hyperpigmentation of skin usually due to estrogen-containing OCPs or pregnancy; worsened with sun exposure; no Tx necessary; can consider stopping OCPs.

- 44M + alcoholic + brought in from the snow in the winter; student Q showed pic of dude's red feet following rewarming and they asked what electrolyte (high or low) we're most worried about → answer = hyperkalemia → alcoholics susceptible to rhabdomyolysis → lysis of cells releases potassium + myoglobin is nephrotoxic and can cause acute tubular necrosis and hyperkalemia.
- 74-year-old man living in Las Vegas + smoker + no other past medical Hx + they show you following image; Dx?



  - Answer = actinic purpura (aka senile or solar purpura). These are flat, purple discolorations caused by damage to dermal collagen and blood vessels due to chronic sun exposure. Commonly appear on the back of the hands and forearms in elderly. Benign; may be mistaken for bruises. NBME Q describes actinic purpura and then asks for mechanism. The answer is "solar elastosis," which is a term that refers to degenerative changes to the dermis due to chronic sun exposure.
- 14F + no past medical history + has irritation on her arms that she finds embarrassing + Q shows you following image; Q asks diagnosis straight-up.



- Diagnosis = keratosis pilaris. Benign bumps + rough patches on the skin due to build-up of excess keratin. The excess Keratin can block hair follicles and pores, causing small, hard bumps. No treatment necessary. Emollients may be used. Not obscure condition I'm mentioning for fun. Shows up on NBME.

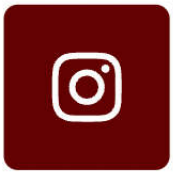
- "Do I need to know the nerve endings in skin and stuff like that?" Annoyingly, they do show up:

Cutaneous nerve endings + what they detect	
Pacinian corpuscles	- Deep pressure and vibrations.
Meissner corpuscles	- Light touch and low-frequency vibrations.
Merkel disc	- Light touch and pressure; provide information about shapes and textures.
Free nerve endings	- Light touch, pain (nociception), temperature change.
Ruffini endings	- Sustained pressure and skin stretching.
Hair follicle receptors	- Hair displacement and gentle touch.



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